

SOC Incident Response Investigation Report

Project Title: Phishing-Led Malware Infection, Credential Compromise, and Possible Data Exfiltration Investigation Using Splunk SIEM.

Analyst Information:

Field	Details
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Role	Junior SOC Analyst
Investigation Date	21-May-2026
Organization	SOC Lab Environment
SIEM Platform	Splunk Enterprise
Investigation Type	Security Incident Response.

Executive Summary:

This investigation focused on identifying and analyzing a suspected phishing-led compromise within a simulated enterprise environment using Splunk SIEM. During the investigation, multiple failed authentication attempts were identified against a corporate user account, followed by a successful login originating from the same external IP address. Shortly afterward, suspicious executable activity, encoded PowerShell execution, suspicious DNS queries, malware detections, and abnormal outbound traffic were observed.

The investigation revealed evidence consistent with:

- Credential compromise
- Possible brute force or credential stuffing attack
- Malware execution
- PowerShell abuse

- Command and Control (C2) communication
- Possible data exfiltration activity
- Phishing-based initial access

Several Indicators of Compromise (IOCs) were identified and correlated across multiple log sources including:

- Windows Security Logs
- Sysmon Logs
- DNS Logs
- Defender Logs
- Network Traffic Logs
- VPN Logs
- Firewall Logs
- Helpdesk/User Reports

The incident was reconstructed through timeline correlation and behavioral analysis.

Investigation Objectives:

The primary objectives of this investigation were:

1. Identify suspicious authentication activity.
 2. Determine whether a compromise occurred.
 3. Investigate malware execution behavior.
 4. Analyze PowerShell activity for malicious indicators.
 5. Investigate DNS activity for Command and Control communication.
 6. Detect signs of possible data exfiltration.
 7. Correlate logs across multiple security sources.
 8. Recommend containment and remediation actions.
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Environment Overview:

Component	Details
Operating System	Windows Environment
SIEM Platform	Splunk Enterprise

Endpoint Monitoring	Sysmon
Endpoint Protection	Microsoft Defender
Network Monitoring	Firewall + Network Traffic Logs
Authentication Source	Windows Security Events
DNS Monitoring	DNS Query Logs

Data Sources Used During Investigation:

Log Source	Purpose
Windows Security Logs	Authentication analysis
Sysmon Logs	Process execution analysis
DNS Logs	Domain query investigation
Defender Logs	Malware detection analysis
Network Logs	Traffic and exfiltration analysis
Firewall Logs	Network connection filtering
VPN Logs	External access analysis
Helpdesk Ticket	User-reported symptoms

Incident Timeline:

Time	Event Description
07:39	Multiple failed login attempts detected for user j.adewale
07:41	Successful login observed from same external IP
07:44	User opened suspicious email attachment
07:45	update_kb5031.exe executed from Temp directory

07:45 Outlook.exe identified as parent process

07:46 Encoded PowerShell execution detected

07:46 Suspicious DNS query to cdn-updates365.com

07:48 Large outbound HTTPS traffic detected

07:49 Defender detected Trojan:Win32/Wacatac.B!ml

07:50 Firewall detected suspicious TCP communication

Task 1:Authentication Investigation:

Objective:

Investigate suspicious login activity and determine whether unauthorized access occurred.

Spl query used: index=soc_lab EventCode=4625 OR EventCode=4624

The screenshot displays the Splunk Enterprise web interface. At the top, a search bar contains the query `index=soc_lab EventCode=4625 OR EventCode=4624`. Below the search bar, the results are shown in a table format. The table has two columns: 'Time' and 'Event'. There are four rows of data, all from May 21, 2026, at 7:41:33 AM. The events are categorized as 'Successful login,NTLM' and 'Unknown user name or bad password,NTLM'. The interface also shows a sidebar with 'INTERESTING FIELDS' and a 'New Search' button.

Time	Event
2026-05-21 07:41:33.462 AM	Successful login,NTLM
2026-05-21 07:40:02.462 AM	Unknown user name or bad password,NTLM
2026-05-21 07:39:51.462 AM	Unknown user name or bad password,NTLM
2026-05-21 07:39:44.462 AM	Unknown user name or bad password,NTLM

Key Findings:

The authentication logs revealed multiple failed login attempts (EventCode 4625) targeting the account: j.adewale

The failed attempts originated from the external IP address: **185.193.88.71**

The same IP address later generated a successful authentication event:

EventCode=4624

Additional observations:

- Authentication Package: NTLM
 - Logon Type: 3 (Network Logon)
 - Same username reused repeatedly
 - Same source IP reused repeatedly
 - Failed logins occurred before successful login
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Analyst Assessment

This pattern strongly indicates:

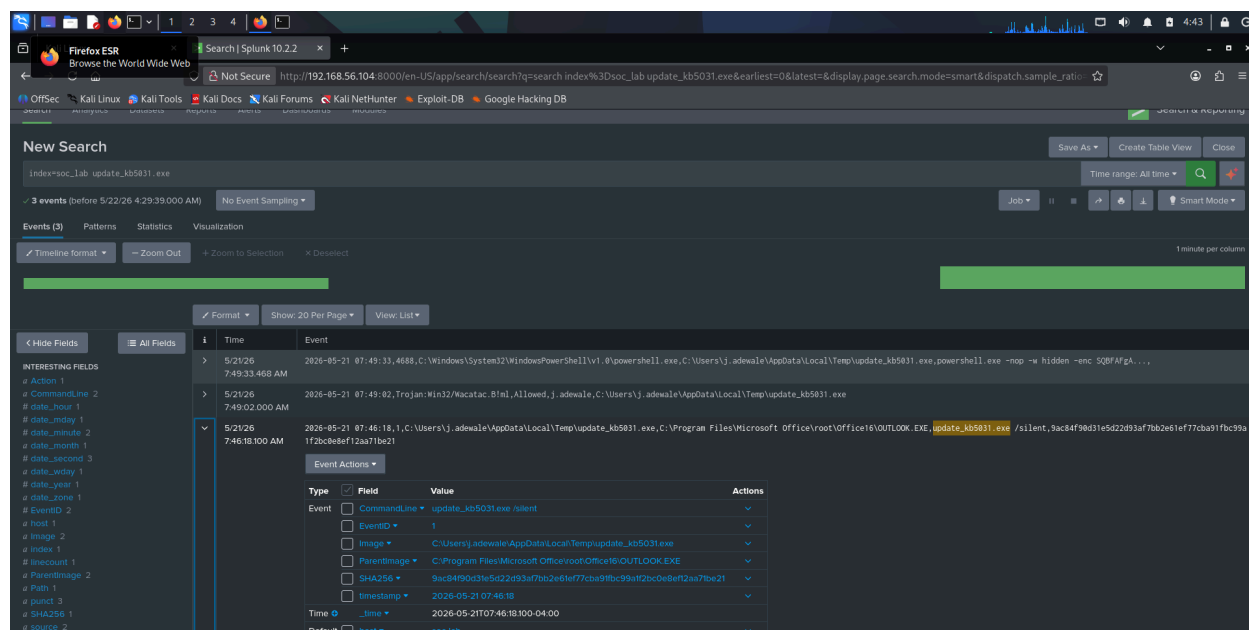
- Brute force activity OR
- Credential stuffing attack

The successful authentication following repeated failures suggests the attacker may have successfully obtained valid credentials.

Task 2: Malware Execution Investigation

Objective

Investigate suspicious process execution activity and identify potential malware behavior.



`index=soc_lab update_kb5031.exe`

Key Findings

The investigation identified suspicious executable activity: **update_kb5031.exe**

The executable was launched from: **`C:\Users\j.adewale\AppData\Local\Temp\`**

Additional findings:

- Parent Process: OUTLOOK.EXE
- Event ID: 1 (Sysmon Process Creation)
- Command-Line Execution Detected
- SHA256 Hash Observed

The executable was launched shortly after the user opened a suspicious email attachment.

Analyst Assessment

Execution of binaries from a Temp directory is highly suspicious, especially when:

- Spawned by Outlook.exe
- Triggered shortly after email interaction
- Associated with malware detections

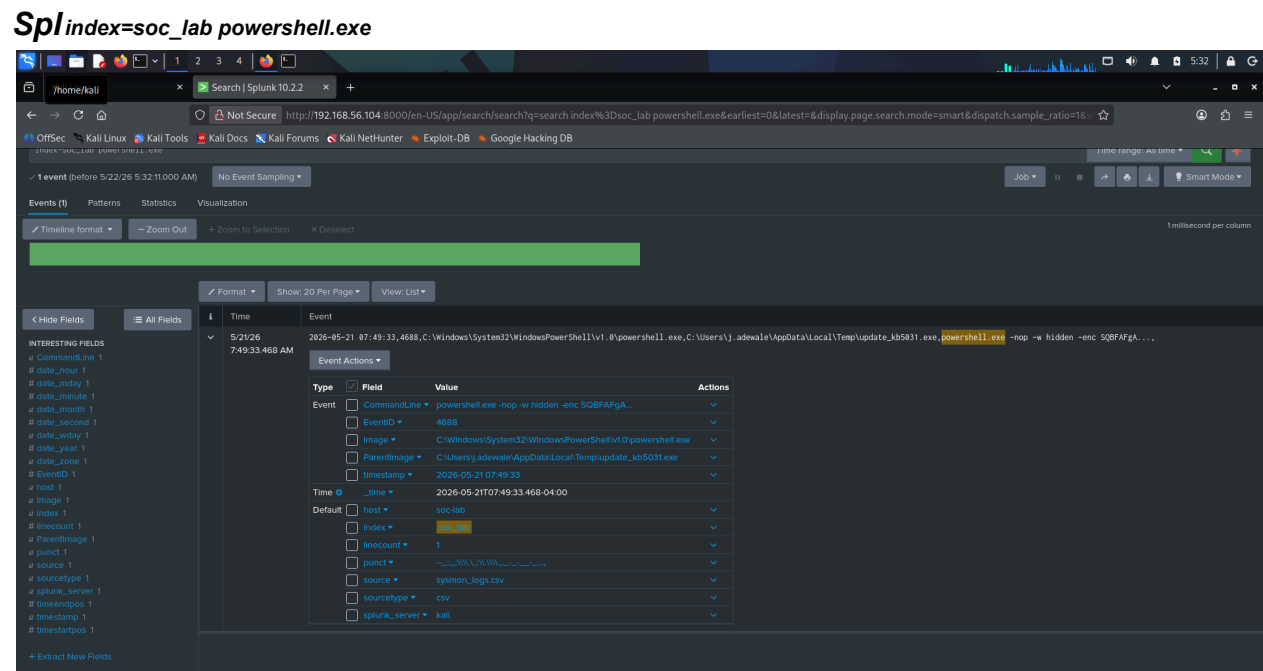
The evidence strongly suggests:

- Phishing-based malware delivery
- Malicious attachment execution
- User compromise through email interaction

Task 3: PowerShell Abuse Investigation

Objective

Investigate suspicious PowerShell activity and determine whether attacker tradecraft was used.



Key Findings:

The investigation identified suspicious PowerShell execution: ***powershell.exe -nop -w hidden -enc***

Observed behaviors included:

- Encoded commands (-enc).

- Hidden window execution (-w hidden)
- No PowerShell profile (-nop)
- Parent process linked to suspicious executable

The PowerShell process was spawned by: ***update_kb5031.exe***

Analyst Assessment

This PowerShell execution was determined to be suspicious because:

- Encoded commands are commonly used by attackers to evade detection.
- Hidden execution indicates stealth behavior.
- No-profile execution avoids defensive controls.
- The parent process originated from a suspicious Temp directory executable.

These behaviors are commonly associated with:

- Malware execution
 - PowerShell abuse
 - Post-exploitation activity
 - Defense evasion techniques
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Task 4: DNS and Command & Control Investigation:

Objective

Investigate suspicious DNS activity and determine possible Command & Control communication.

Spl: index=soc_lab cdn-updates365.com

Key Findings:

DNS logs showed repeated queries to the domain: **cdn-updates365.com**

Observations:

- Queries originated from compromised host FIN-WS224
- Repeated DNS lookups observed
- Queries occurred shortly after malware execution
- Domain naming pattern imitated legitimate Microsoft infrastructure

Analyst Assessment

The domain appears suspicious because:

- It mimics trusted update infrastructure.
- The compromised host repeatedly queried the domain.
- Queries occurred immediately after malware execution.
- The activity aligns with malware beaconing behavior.

This may indicate:

- Command & Control communication
 - Malware callback attempts
 - Secondary payload retrieval
-

Task 5: Network Traffic and Data Exfiltration Investigation

Objective:

Investigate outbound network activity for signs of exfiltration.

Splunk Query Used

```
index=soc_lab BYTES_OUT>10000000
```

Key Findings

Large outbound HTTPS traffic was detected from: **10.10.44.23 to:45.77.91.203**

Observations:

- Multiple outbound transfers detected
 - Large byte transfers observed
 - Traffic occurred over TCP/443
 - Repeated communication occurred within seconds
 - Activity occurred shortly after malware execution
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Analyst Assessment

The outbound traffic was determined to be suspicious because:

- The host was previously identified as compromised.
- Large outbound encrypted traffic was observed.
- The destination was an external IP address.
- Timing aligned with malware activity.

This behavior strongly suggests:

- Command & Control communication OR
- Possible data exfiltration

Indicators of Compromise (IOCs):

Malicious IP Addresses:

IOC Type	Value
External IP	185.193.88.71
External IP	45.77.91.203
VPN Source IP	197.210.77.19

Suspicious Domain:

IOC Type	Value
Domain	cdn-updates365.cm

Malicious File:

IOC Type	Value
Executable	update_kb5031.exe
ThreatName	Trojan:Win32/Wacatac.B!ml

Suspicious PowerShell Indicators:

IOC Type	Value
Encoded PowerShell	-enc
Hidden Execution	-w hidden

No Profile

-nop

MITRE ATT&CK Mapping:

Tactic	Technique
Initial Access	Phishing.
Credential Access	Brute Force / Credential Stuffing.
Execution	PowerShell.
Defense Evasion	Obfuscated Files or Information.
Command and Control	Application Layer Protocol.
Exfiltration	Exfiltration Over Web Services.

Containment Actions Recommended:

The following containment actions were recommended:

1. Isolate affected endpoint FIN-WS224 from the network.
2. Disable or reset the compromised user account: j.adewale
3. Block the malicious IP addresses and suspicious domain: cdn-updates365.com
4. Terminate malicious PowerShell and suspicious processes.
5. Quarantine and remove: update_kb5031.exe
6. Re-enable Microsoft Defender and perform a full endpoint scan.
7. Review the environment for lateral movement or additional compromised systems.
8. Preserve logs and forensic evidence for further investigation.